# NRR-Guarantee: Bounded Commit/Defer Guarantee Surfaces and Explicit Downstream Boundaries

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**Abstract**

This manuscript sets the current downstream `Guarantee` surface for the Non-Resolution Reasoning (NRR) program. Its purpose is to close the next paper as a bounded claim layer rather than as a broad universality paper. Under explicit provider, family, horizon, and observability conditions, the current manuscript integrates one supported claim surface, one downstream-acceptance boundary surface, one side-topic supporting surface, and one later-horizon boundary surface: frozen early ambiguity-turn resolution, downstream acceptance under the inherited mainline comparison contract, repaired first-step side-topic/quarantine workstart, and later-horizon relation-preservation failure reporting. On the current tested-provider surface, the early ambiguity-turn read moves from `0 / 24` to `24 / 24` on both providers for `resolution_attempt_present` and `appropriate`, and from `8 / 24` to `24 / 24` for `unilateral_commit_absent`; the repaired downstream-acceptance boundary supports bounded `preacceptance_rework_burden_absent` improvement under matched `baseline vs Energy policy` conditions (both provider slices move from `0 / 24` to `24 / 24`, while Gemini `accepted_direction_followthrough_present` drops from `13 / 24` to

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10 / 24), the auxiliary repaired NRR vs Ablation read remains mixed (Anthropic = inconclusive, Gemini = support; Anthropic Baseline remains stronger than Anthropic NRR on appropriate), and the repaired first-step side-topic surface reaches 12 / 12 on both providers. That side-topic surface is carried only as bounded supporting evidence because it depends on a local repaired baseline vs revised Energy policy comparison rather than on the paper-level mainline baseline vs Energy policy contract. The result is a narrow downstream paper surface: one supported guarantee claim plus explicit downstream, side-topic, and later-horizon limits. Across those surfaces, the honest read is therefore non-uniform: the downstream-acceptance boundary remains provider-sensitive beyond its shared burden-removal gain, and the later-horizon material remains boundary-only rather than an additional supported guarantee.

# 1 Introduction

NRR is not an anti-LLM framework. NRR does not replace standard LLM use. NRR optimizes when to commit and when to defer, under explicit conditions.

Related work in mixed-initiative and task-oriented dialogue has long examined adaptive initiative control and goal-oriented interaction management [8, 10], while clarification-focused lines study how a system should ask follow-up questions when a user's request remains ambiguous or underspecified [9, 11, 12]. Grounding-oriented work further emphasizes that dialogue systems must recover collaboratively when interpretation is not yet fixed [13, 14]. More recent LLM-centered work has also revisited clarification under query ambiguity and abstention or withholding under uncertainty [15, 16, 17]. Work on verified AI and runtime assurance frames a complementary question: what scoped assurances can be justified under explicit assumptions, monitors, and failure boundaries rather than by unconstrained behavioral optimism [18, 19]. The present manuscript uses Guarantee in that bounded verification/assurance sense: not as a theorem-level or provider-universal correctness claim, but as a conditioned claim surface whose provider scope, horizon scope, observability scope, and failure boundaries are made explicit and auditable.

Within the NRR series, Core and Phi fixed the upstream position-dependent identity and state-mapping surfaces [1, 2]. IME, Transfer, and Coupled then constrained the interface, transfer, and dependency surfaces [3, 4, 5]. Integrated paper7 fixed the manuscript-facing delayed-commitment comparison and boundary-honesty surface, and Energy fixed the current calibration/control handoff that the present manuscript now

inherits as its direct upstream spine [6, 7].

The present manuscript instead treats `Policy`, `Dynamics`, and `Policy-Verification` as manuscript-facing supporting layers for a bounded `Guarantee` read rather than as independent completion targets or a serial upstream chain. Accordingly, its purpose is to state only those downstream surfaces that remain supportable when the manuscript-facing dependency spine is kept at `paper7` → `Energy` → `Guarantee` and when provider, family, horizon, observability, and failure-boundary limits are made explicit.

The Core-to-Guarantee program goal itself is not being rewritten here. What the series must establish from `Core` through `Guarantee` remains the same: a framework grounded in position-dependent identity as a first principle, closed as an engineering program of conditional utility through implementation, quality measurement, operational control, and verification. The current *non-evaluative principle* read reorganizes that same program; it does not replace the top-level goal. In this manuscript, the practical use of that read is to tighten closure discipline around a narrow notion of 'evaluation', namely full branchwise comparative evaluation during retention.

The present manuscript line is intentionally narrow. It states the currently supported bounded claim surface without promoting any unverified provider-universal, family-universal, or long-horizon statement.

**Series Consistency Statement:** Unsupported interpretations are never injected at a fixed evidence state, hypothesis-set expansion is evidence-gated, and unobserved possibilities remain explicitly unresolved. In this paper, the evidence-source clause is restricted to the frozen downstream support layers and the directly imported observability surfaces named for the current `Guarantee` manuscript line.

**Series Extensibility Statement:** NRR is a derivation-and-evaluation framework, not a closed recipe. The currently evaluated policy/operator set is treated as an evaluated set under tested conditions, not as an exhaustive set. The aim of the present manuscript is not to imply that the current runs exhaust the guarantee question, but to fix a reusable test surface: one bounded supported claim, explicit supporting and boundary surfaces, explicit comparison rules, provider-sliced reporting, and visible failure boundaries that later readers can rerun, challenge, and extend under new providers, new families, and longer horizons.

## 2 Contributions

Under the currently fixed provider, family, horizon, and observability bounds, this manuscript makes four bounded contributions:

1. it establishes bounded early ambiguity-turn support for non-unilateral priority resolution under matched `baseline vs Energy policy` conditions;

2. it carries downstream acceptance as an explicit boundary surface: the mainline `baseline vs Energy policy` comparison supports shared `preacceptance_rework_burden_absent`, while visible provider-sliced residuals prevent pair-wide downstream-quality closure;

3. it carries repaired first-step side-topic / quarantine evidence as a bounded supporting surface under matched `baseline vs revised Energy policy` conditions, with the main incremental lift concentrated in `C_incident_response` and without upgrading that local repaired comparison into a second manuscript supported claim;

4. it keeps later-horizon degradation and unsupported extensions explicit as failure-boundary reporting and reusable rerun surfaces rather than inflating them into broader guarantee language.

## 3 Program Position And Scope

This paper sits at the end of the current manuscript-facing downstream spine:

$$\texttt{paper7} \rightarrow \texttt{Energy} \rightarrow \texttt{Guarantee}$$

The manuscript-facing dependency split is fixed for interpretation. Integrated `paper7` supplies the comparison-and-boundary-honesty upstream surface. `Energy` supplies the calibration/control handoff and the current imported evidence surface. `Policy` supplies the minimal state/action specification. `Dynamics` supplies the hold/degrade/break vocabulary over later checkpoints. `Policy-Verification` supplies the prospective matched- condition validation surface. `Guarantee` then carries the bounded supported, supporting, and boundary surfaces that remain after those inputs are imported under explicit conditions. Figure 1 summarizes that manuscript-facing dependency structure.

This means the manuscript has a double obligation. It must absorb the already fixed supporting layers without reopening their ownership boundaries, and it must state only those guarantee-shaped claims that remain supportable after explicit restriction of provider, family, horizon, observability, and failure boundary.

The current organizing read is likewise narrow. The *non-evaluative principle* is used here only as a local way to organize what counts as satisfactory closure for this manuscript, not as a theorem-level claim that supersedes the Core-to-Guarantee objective itself. In that narrower sense, `Guarantee` is the downstream line where the bounded supported, supporting, and boundary surfaces must be stated without inflating them into unrestricted runtime branchwise comparison.

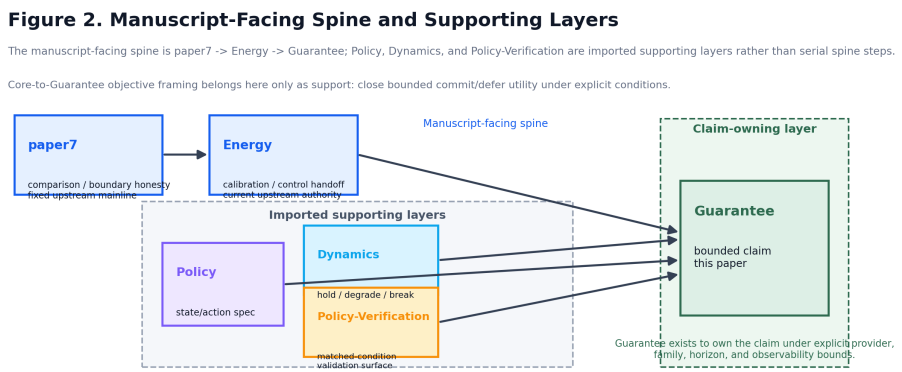


Figure 1: Current manuscript-facing dependency structure. Integrated `paper7` and `Energy` remain the manuscript-facing upstream spine, while `Policy`, `Dynamics`, and `Policy-Verification` are imported supporting layers rather than serial spine steps. `Guarantee` carries the current bounded supported and supporting, and boundary surfaces.

# 4 Supporting-Layer Inputs

## 4.1 Policy Input

The current policy layer fixes the downstream state vocabulary (`active_objective`, `suspended_objective`, `accepted_direction`, `unresolved_relation`, `quarantine`, `local_relation_block`) together

with the minimal commit/defer action set. The `Guarantee` line inherits this vocabulary as its operating language and does not redesign it locally.

That vocabulary is not ornamental. The early ambiguity-turn surface is the cleanest current instance of `defer_with_resolution`: no content work should begin before a direction is accepted, and no unilateral priority choice should be made on the user's behalf. The downstream acceptance surface is the current instance of `commit_to_accepted_direction` in which the honest carry-forward read is bounded support for `preacceptance_rework_burden_absent` under the mainline comparison contract, with Gemini residuals still visible beyond that shared gain rather than broad downstream closure. The repaired first-step side-topic surface is the clearest first-step instance of `accepted_direction` becoming `active_objective` while the prior line is held as `suspended_objective` under `quarantine`, but in the current paper it is carried only as a supporting surface because that read depends on a local repaired comparison rather than on the manuscript's inherited mainline comparison contract. The later-horizon boundary then marks the point where `unresolved_relation` handling weakens and `local_relation_block` is no longer treated conservatively. For the current downstream line, preserving an unresolved relation should also be read in a narrower but more practical sense than mere recommendation avoidance. Once a branch has already been accepted, failure can still occur if the next assistant move silently compresses still-live user-side constraint or satisfaction structure into one asserted local reading. The manuscript does not expand that point into a general hidden-needs thesis. It uses it only to mark a bounded downstream failure mode: advancing as though the user's governing criterion were already fixed when the visible interaction state does not yet support that fixation.

## 4.2 Dynamics Input

The current dynamics layer contributes mechanism vocabulary only. Its role is to name where the inherited policy state holds, weakens, or breaks across later checkpoints. In the current branch, the key motifs are *Stable Resolution*, *Clean Quarantine Hold*, *Useful Continuation Hold*, *Relation-Preservation Hold*, *Premature Relation Collapse*, *Repeated-Starter Attractor*, *Generic Stall*, and *Quarantine Leak*.

Using that vocabulary makes the manuscript read as one bounded progression rather than as three disconnected pilot wins. The ambiguity-turn result is the clearest instance of *Stable Resolution*. The repaired first-step side-topic supporting surface is the clearest instance of

*Clean Quarantine Hold*. The downstream acceptance result remains narrower: under the manuscript's inherited `baseline vs Energy policy` comparison, the shared supported behavior is bounded removal of `preacceptance_rework_burden_absent`, but Gemini `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`, and the auxiliary repaired `NRR vs Ablation` read remains mixed, with Anthropic `Baseline` still stronger than Anthropic `NRR` on `appropriate`. That surface should not yet be collapsed into uniform *Useful Continuation Hold*. Because the repaired side-topic read depends on a local repaired comparison, it remains supporting evidence rather than a second supported claim. The later-horizon boundary is best read as weakening *Relation-Preservation Hold* before *Premature Relation Collapse* appears, with more local residual forms such as *Generic Stall* or *Repeated-Starter Attractor* where the template/provider slice warrants them.

### 4.3 Policy-Verification Input

The current policy-verification layer fixes the protocol rules that any prospective `baseline vs Energy policy` comparison must satisfy before guarantee text is allowed. This includes matched-condition comparison, pre-frozen reporting surfaces, narrow family selection, explicit horizon bands, and main-text reporting of both gains and failure boundaries.

### 4.4 Experimental Conditions

The carried surfaces in this manuscript are reported as frozen matched-condition row totals rather than as claims of independent stochastic sampling. The early ambiguity-turn pilot uses temperature `0.0`, three templates, and eight reps per provider/arm, so each provider slice is reported over `24` rows at one early decision boundary. The downstream-acceptance pilot also uses temperature `0.0`, the same three templates, and four reps per acceptance variant across two acceptance variants, again yielding `24` rows per provider slice at the first accepted-direction boundary. The repaired first-step side-topic / quarantine pilot uses temperature `0.0`, the same three templates, `accept_side_first` only, and four reps per provider/arm, yielding `12` rows per provider slice at `turn5`. The later-horizon boundary then reports later checkpoints `turn7`, `turn9`, and `turn11` over `12` episodes per provider in a separate frozen `priority-resolution horizon-stability` family that reads the later continuation of the carried side-topic line. The numerators and denominators in the manuscript should therefore be read

as fixed-condition row counts under frozen prompts, models, and templates, not as provider-universal frequency estimates.

## 5 Claim, Supporting-Surface, And Boundary Discipline

Any supported claim, supporting surface, or explicit boundary surface in this paper must disclose a bounded reporting surface with the following fields:

1. provider scope
2. family scope
3. horizon scope
4. observability scope
5. comparison scope
6. supported behavior
7. explicit failure boundary
8. unsupported extension

These fields define a shared disclosure schema, not one uniform evidentiary contract. Supported claims, supporting surfaces, and matched-condition boundary surfaces must fill all eight fields explicitly. The later-horizon boundary is instead carried as a diagnostic one-arm later-checkpoint degradation surface: it still states provider, family, horizon, observability, comparison, supported behavior, explicit failure boundary, and unsupported extension, but its comparison scope is explicitly a diagnostic one-arm later-checkpoint read rather than a matched-comparison fill, and its supported-behavior field remains boundary-only rather than an additional supported claim. No carried scientific surface is valid if the fields required by its evidentiary contract are omitted. This paper therefore inherits the current schema rule that the claim surface must be chosen before outcome inspection and must remain narrow enough to prevent provider-universal, family-universal, or long-horizon overreach. Table 1 summarizes the supported, supporting, and boundary surfaces under that disclosure schema, and Figure 2 shows how those surfaces relate across the current manuscript-facing progression.

| Unit | Scope | Supported behavior | Boundary kept explicit |
|---|---|---|---|
| Early ambiguity-turn | Tested provider pair, frozen priority-resolution family, early decision boundary | Non-unilateral priority resolution | No downstream acceptance, side-topic/quarantine, or later-horizon extension |
| Downstream acceptance boundary | Tested provider pair, priority-resolution family, first accepted-direction boundary | Bounded support for preacceptance burden removal under matched baseline vs Energy-policy conditions | Keep Gemini residuals and the auxiliary repaired read explicit; do not collapse into pair-wide or provider-universal downstream-guarantee language |
| Side-topic/quarantine supporting surface | Tested provider pair, repaired side-topic family, first accepted side-topic workstart checkpoint | Clean first-step side-topic workstart under explicit side-first acceptance on a local repaired comparison | Keep it as supporting evidence only; do not upgrade the local repaired comparison into a second manuscript supported claim or later-horizon stability read |
| Later-horizon boundary | Tested provider pair, priority-resolution horizon-stability family, later checkpoints `turn7`/`turn9`/`turn11`, diagnostic one-arm later-checkpoint read | Boundary-only read: relation preservation weakens before suspended-main quarantine breaks | Keep later-horizon as diagnostic boundary reporting only; no matched-comparison fill, no fourth supported claim, and no provider-universal or long-horizon extension |

Table 1: Current supported, supporting, and boundary surfaces in `Guarantee`.

## 5.1 First Supported Claim Unit

The first explicit supported claim unit in the current manuscript is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the frozen `priority-resolution` ambiguity-turn family only
- horizon scope: the early decision boundary only
- observability scope: `resolution_attempt_present`, `unilateral_commit_absent`, and `appropriate`
- comparison scope: matched baseline vs Energy-policy under the frozen early ambiguity-turn pilot conditions
- supported behavior: non-unilateral priority resolution
- explicit failure boundary: no downstream acceptance, side-topic/quarantine, or later-horizon closure is established by this fill
- unsupported extension: no provider-universal, long-horizon, or broad downstream-quality claim is authorized

In manuscript form, the bounded claim is:

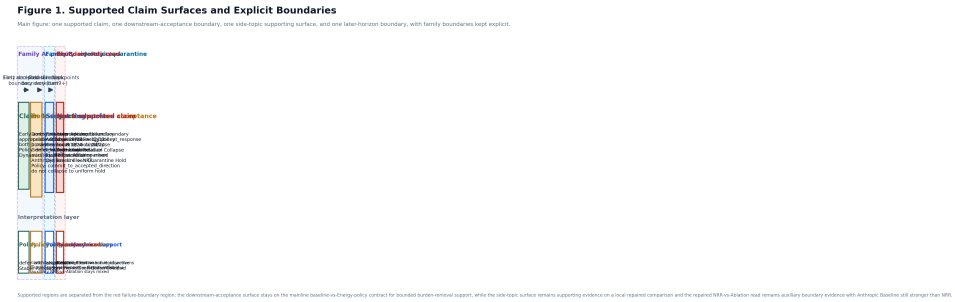


Figure 2: Supported claim progression, supporting evidence, and explicit boundary surfaces. Claim 1 and the downstream-acceptance boundary occupy different horizon points within the same `priority-resolution` family, with the downstream-acceptance surface marking the first accepted-direction boundary inside that family, while the side-topic/quarantine read belongs to a separate repaired family and is carried only as supporting evidence. The later-horizon region is a separate `priority-resolution horizon-stability` boundary read over the later continuation of that carried line rather than an additional supported claim.

> Under the tested provider pair, for the frozen `priority-resolution` ambiguity-turn family, at the early decision boundary, and on the directly scored fields `resolution_attempt_present`, `unilateral_commit_absent`, and `appropriate`, the Energy-policy arm is supported relative to matched baseline only for non-unilateral priority resolution. This support excludes downstream acceptance, side-topic/quarantine, and later-horizon behavior, and it does not extend to provider-universal or broad downstream-quality claims.

## 5.2 Downstream-Acceptance Boundary Surface

The current downstream-acceptance boundary surface in the manuscript is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the carried `priority-resolution` family only
- horizon scope: the first accepted-direction boundary inside that carried family only

- observability scope: `accepted_direction_followthrough_present`, `rejected_direction_intrusion_absent`, `preacceptance_rework_burden_absent`, and `appropriate`
- comparison scope: matched baseline vs Energy-policy under the frozen downstream-boundary pilot conditions, with reviewed `NRR vs Ablation` retained only as an auxiliary repaired boundary comparator
- supported behavior: bounded support for removal of preacceptance rework burden under the first accepted-direction downstream boundary
- explicit failure boundary: the surface does not close as pair-wide downstream quality, provider-universal continuation support, explicit quarantine closure, or later-horizon continuation stability; Gemini `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`, and the auxiliary repaired read remains mixed
- unsupported extension: no pair-wide downstream-acceptance guarantee, explicit quarantine closure, or later-horizon stability claim is authorized, and no manuscript claim that the auxiliary repaired `NRR vs Ablation` read replaces the paper-level comparison contract

In manuscript form, the bounded boundary read is:

> Under the tested provider pair, within the carried `priority-resolution` family, at the first accepted-direction boundary, and on the directly scored fields `accepted_direction_followthrough_present`, `rejected_direction_intrusion_absent`, `preacceptance_rework_burden_absent`, and `appropriate`, the matched `baseline vs Energy policy` surface supports only bounded removal of preacceptance rework burden rather than pair-wide downstream-quality closure: both provider slices reach `24 / 24` on `preacceptance_rework_burden_absent`, but Gemini `accepted_direction_followthrough_present` simultaneously drops from `13 / 24` to `10 / 24` on the same mainline comparison surface. The reviewed `NRR vs Ablation` read remains auxiliary boundary evidence only: Gemini reaches support there, Anthropic remains inconclusive, and the Anthropic-side auxiliary comparison keeps `Baseline` stronger than `NRR` on `appropriate`. This surface is therefore carried forward only as a bounded downstream-boundary result, not as a pair-wide downstream guarantee, explicit quarantine closure, or later-horizon stability claim.

The auxiliary repaired read uses a narrower direct trace than the mainline protocol-era fields: `accepted_branch_followthrough_present` maps to `accepted_direction_followthrough_present`, `reopened_`

`choice_absent` plus `alternative_branch_push_absent` jointly trace the manuscript's rejected-direction non-intrusion requirement, and that mapping is kept explicit in the bundled closure import note rather than treated as a one-to-one field replacement.

### 5.3 Side-Topic / Quarantine Supporting Surface

Because this surface depends on a local repaired `baseline vs revised Energy policy` comparison rather than on the paper-level mainline `baseline vs Energy policy` contract, it is carried here as a bounded supporting surface and not as a second manuscript supported claim unit.

The repaired first-step side-topic / quarantine supporting surface in the current manuscript is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the repaired first-step side-topic / quarantine family only
- horizon scope: the first accepted side-topic workstart checkpoint only
- observability scope: `chosen_side_direction_workstart_present`, `suspended_main_reference_absent`, `main_direction_intrusion_absent`, and `appropriate`
- comparison scope: matched baseline vs revised Energy-policy under the repaired first-step side-topic / quarantine pilot conditions
- supported behavior: clean first-step side-topic workstart under explicit side-first acceptance
- explicit failure boundary: the gain is incremental rather than a rescue from total baseline failure, and the main lift over baseline is concentrated in `C_incident_response`; because this read uses a local repaired comparison rather than the inherited paper-level contract, it is carried as supporting evidence rather than as a manuscript supported claim unit
- unsupported extension: no provider-universal side-topic commitment, uniform template lift, later- horizon side-topic stability claim, or upgrade to a second manuscript supported claim is authorized

In manuscript form, the bounded supporting read is:

Under the tested provider pair, for the repaired first-step side-topic / quarantine family, at the first accepted side-topic workstart checkpoint, and on the directly scored fields `chosen_side_direction_workstart_present`, `suspended_main_reference_absent`, `main_direction_intrusion_absent`, and `appropriate`, the revised Energy-policy arm is supported relative to matched baseline only for clean first-step side-topic workstart under explicit side-first acceptance. This surface is carried only as bounded supporting evidence because its comparison contract is the repaired local `baseline vs revised Energy policy` surface rather than the manuscript's mainline `baseline vs Energy policy` contract. It excludes provider-universal side-topic commitment, uniform template lift, and later-horizon side-topic stability, and it remains bounded by the fact that the main incremental lift over baseline is concentrated in `C_incident_response`.

## 6 Evidence Import Surface

The scientific argument in this manuscript draws on three bundled evidence layers. Fixed supporting-layer texts provide the inherited program-design, policy, dynamics, protocol, and historical handoff vocabulary that is imported without reopening ownership locally. Repo-local analysis notes then fix the current manuscript-facing reads for the early ambiguity-turn claim, the downstream-acceptance boundary, the repaired first-step side-topic / quarantine supporting surface, the later-horizon boundary, the policy/dynamics interpretation layer, and the current upstream spine interpretation. An imported reviewed readout surface preserves the repaired `NRR vs Ablation` downstream comparison as auxiliary boundary evidence only.

Detailed artifact routing, filenames, and review-package entrypoints are kept outside the scientific body in `reproducibility.md` and the current supporting-layer evidence map. The manuscript body uses those bundled inputs to state the bounded scientific read rather than to reproduce the repository manifest inline.

This split is deliberate. The support bundle carries fixed upstream dependency texts and historical provenance; it is not by itself the manuscript-facing proof surface for every later bounded surface. In particular, the older downstream- acceptance memo preserved in `supporting_layer_sources/current/` remains a pre-repair historical read. The current downstream-acceptance boundary surface is instead anchored by the repo-local closure import note, the repo-local boundary-surface note, the directly shipped baseline-vs-Energy comparison memo with its four run-annotation CSVs, and the imported reviewed auxiliary repaired surface listed above. The shipped bundle now also includes the repaired side-topic

comparison memo together with its four run-annotation CSVs, and the later-horizon two-provider memo together with the provider-scored read memos and run-annotation CSVs needed to inspect that boundary one layer below the manuscript-facing drafting memos.

### 6.1 First Evidence Import

The strongest current entry point for the `Guarantee` line is the frozen early ambiguity-turn pilot. Under matched baseline vs Energy-policy conditions, both tested provider slices moved from `0 / 24` to `24 / 24` on `resolution_attempt_present`, from `8 / 24` to `24 / 24` on `unilateral_commit_absent`, and from `0 / 24` to `24 / 24` on `appropriate`. Baseline failure modes remained visible as either ambiguity-preserving but unresolved responses or unilateral/content-before-resolution behavior. The carry-forward read is therefore that the Energy-policy surface supports non-unilateral priority resolution at the early decision boundary on the tested provider pair. This is a bounded early-control claim only; it does not by itself establish downstream followthrough, quarantine behavior, later-horizon stability, or provider-universal validity.

### 6.2 Second Evidence Import

The downstream-acceptance boundary is the current second `Guarantee` entry point. The inherited paper-level comparison contract stays on matched `baseline vs Energy policy` conditions, where both tested provider slices move from `0 / 24` to `24 / 24` on `preacceptance_rework_burden_absent`. That shared gain is the current supported downstream-boundary behavior. The residual remains visible in the same mainline surface: Gemini still shows partial first accepted-direction followthrough on part of the tested rows, so the claim does not close as pair-wide downstream-quality support. The imported reviewed readout then keeps a narrower auxiliary repaired comparison visible directly inside this repository: on reviewed `NRR vs Ablation`, Gemini reaches support, Anthropic remains inconclusive, and the same reviewed memo keeps Anthropic `Baseline` stronger than Anthropic `NRR` on `appropriate`. The truthful carry-forward read is therefore bounded support for preacceptance burden removal under the mainline comparison, with the repaired `NRR vs Ablation` surface used only as auxiliary boundary evidence. This surface does not yet license pair-wide downstream guarantee language, explicit quarantine closure, later-horizon stability claims, or any claim that the auxiliary repaired read replaces the paper-level comparison

contract.

### 6.3 Third Evidence Import

The third manuscript-facing `Guarantee` entry point is the repaired first-step side-topic / quarantine supporting surface. Under matched baseline vs revised Energy-policy conditions, the revised arm reached `appropriate = 12 / 12` in both provider slices, but the honest read is incremental rather than global: baseline was already fairly strong on `A_trip_planning` and `B_hiring_packet`, and the main lift over baseline concentrated in `C_incident_response` (`1 / 4 -> 4 / 4` on Anthropic and `0 / 4 -> 4 / 4` on Gemini). Because this surface sits on a local repaired `baseline vs revised Energy policy` comparison rather than on the paper-level mainline contract, the carry-forward read is a bounded first-step side-topic workstart supporting surface under explicit side-first acceptance, not a second manuscript supported claim and not a universal side-topic/quarantine or later-horizon stability claim.

**Prospective-Killer Bridge Note** A separate prospective-killer pilot on the repaired English bridge surface does not enter this paper as a supported `Guarantee` claim unit or as a boundary surface. Its narrower manuscript role is to show that prompt-level `NRR-style` control can materially improve multi-turn behavior under the tested English scenario protocol while still leaving an explicit later- followthrough residual, which makes the pilot useful here only as a bounded bridge note rather than as a new guarantee result. In the repaired full run, overall `final_success_rate` moves from `0.802` to `0.904` and `NRR` `final_success_rate` moves from `0.472` to `0.769`; the main gain is in `downstream_acceptance`, where `DA / nrr` moves from `0.083` to `0.861`, while `EA / nrr` remains strong (`0.861 -> 0.889`) and `LF / nrr` improves only partially (`0.472 -> 0.556`). The explicit carried-over residual `LF_002 / gemini / nrr` remains inside the scored set. The associated `Gemini-only logprobs` read remains mechanistic sidecar evidence only. Its role in this manuscript is therefore a secondary bounded bridge note inside the evidence-import discussion rather than a main claim unit, a formal carried surface, or proof that a non-softmax `NRR-native` architecture is already established.

**Formal Carried Surfaces** The formal scientific surfaces carried by this manuscript are therefore three comparison-backed carried surfaces plus one diagnostic later-horizon boundary:

- Early ambiguity-turn supported claim: tested provider pair, early decision boundary, and direct observability on `resolution_attempt_present`, `unilateral_commit_absent`, and `appropriate`.
- Downstream-acceptance boundary: tested provider pair, first accepted- direction downstream boundary, and direct observability on `accepted_direction_followthrough_present`, `rejected_direction_intrusion_absent`, `preacceptance_rework_burden_absent`, and `appropriate`.
- First-step side-topic / quarantine supporting surface: tested provider pair, first accepted side-topic workstart checkpoint, and direct observability on `chosen_side_direction_workstart_present`, `suspended_main_reference_absent`, `main_direction_intrusion_absent`, and `appropriate`; this surface remains supporting evidence because it depends on a local repaired `baseline vs revised Energy policy` comparison.
- Later-horizon failure boundary: tested provider pair, measured later checkpoints especially `turn9`, and relation-preservation failure surfaces under a diagnostic one-arm later-checkpoint read.

The prospective-killer pilot remains outside this formal surface list and is carried only as a secondary bounded bridge note.

### 6.4 Policy/Dynamics Interpretation Layer

Taken together, these carried surfaces define a bounded progression rather than a universal recipe. They share one disclosure schema, but not one uniform evidentiary contract: the first three surfaces are comparison-backed, while the later-horizon boundary is a diagnostic later-checkpoint read. At the earliest boundary, the supported behavior is `defer_with_resolution` and the corresponding dynamics motif is *Stable Resolution*. At the downstream acceptance boundary, the supported behavior is bounded burden removal under matched `baseline vs Energy policy` conditions, but Gemini still retains first-followthrough residuals and the auxiliary repaired `NRR vs Ablation` read remains mixed, so this surface cannot yet be read as pair-wide *Useful Continuation Hold*. At the repaired first-step side-topic supporting surface, the carried behavior is a clean transition from `accepted_direction` into `active_objective` with `suspended_objective` still held in quarantine, which corresponds to *Clean Quarantine Hold*. Because that surface depends on a local repaired comparison, it remains supporting evidence rather than

the paper's second supported claim. The later-horizon limit then becomes easier to state precisely: the first broad break is not simple suspended-main reactivation but weakening *Relation-Preservation Hold* until *Premature Relation Collapse* appears. Table 2 summarizes the provider-specific residual structure that keeps the current `Guarantee` claims bounded.

## 7 Later-Horizon Failure Boundary

The current later-horizon boundary surface in the manuscript is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the frozen `priority-resolution horizon-stability` family only, separate from the repaired first-step side-topic/quarantine family and used to read the later continuation of the carried side-topic line
- horizon scope: later checkpoints `turn7`, `turn9`, and `turn11` only, with `turn9` the first shared clean failure checkpoint
- observability scope: `chosen_side_direction_progress_present`, `user_detail_use_present`, `unresolved_relation_preserved_or_narrowly_checked`, `suspended_main_reference_absent`, `starter_question_repeat_absent`, and `appropriate`
- comparison scope: not a matched comparison fill; diagnostic one-arm later-checkpoint read over the provider-scored annotation tables for the active later-horizon family
- supported behavior: boundary-only detection that relation preservation weakens before suspended- main quarantine breaks, with the limited safe hold that the first side-topic switch can survive into the next checkpoint
- explicit failure boundary: later degradation remains real, template-dependent, and provider-sensitive; the clearest shared failure surface is `C_incident_response` at `turn9`, where both providers collapse the unresolved relation by prematurely committing an ordering between `impact` and `root cause`

- unsupported extension: no matched-comparison claim, no fourth supported `Guarantee` claim, no provider-universal or family-universal degradation depth, no long-horizon stability guarantee, and no claim that prompt-layer control alone secures stable `turn11` behavior

  In manuscript form, the bounded later-horizon boundary read is:

> Under the tested provider pair, for the frozen `priority-resolution horizon-stability` family, kept separate from the repaired first-step side-topic/quarantine family but used to read the later continuation of the carried side-topic line, at later checkpoints `turn7`, `turn9`, and `turn11`, and on the directly scored later-horizon observability fields listed above, the current later-horizon surface is carried only as a diagnostic one-arm boundary read, not as a matched-comparison fill. The limited safe hold is that the first side-topic switch can survive into the next checkpoint while suspended-main quarantine remains visible, but later degradation remains real, template-dependent, and provider-sensitive. The clearest shared failure surface is the `C_incident_response` template at `turn9`, where both providers collapse the still unresolved relation by prematurely committing an ordering between `impact` and `root cause`. The correct read is therefore that later-horizon instability emerges through weakening relation preservation before suspended-main quarantine itself breaks. This later-horizon material is a boundary on the current supported surfaces only: it does not authorize a matched-comparison claim, a fourth supported `Guarantee` claim, provider-universal degradation depth, or long-horizon stability language.

## 8 Reproducibility And Repository Surface

The tracked repository surface for this manuscript is intentionally narrow and repo-local:

- a current TeX source
- a current PDF snapshot
- the current figure assets used by that TeX source
- a supporting-layer evidence map
- bounded claim, supporting-surface, and boundary drafting memos
- an imported reviewed downstream-acceptance readout surface for the current downstream boundary
- a secondary prospective-killer bridge note

- a later-horizon failure-boundary memo
- a policy/dynamics manuscript-integration memo
- a bundled supporting-layer source surface
- stable build and verification wrappers

The repository therefore exposes both the manuscript-facing claim surface and the specific imported source layers used to justify it. Checksum verification executes on the tracked current artifact set, and the documented build wrapper produces a current PDF in this environment through a source-rendered fallback when cached TeX resources are incomplete. Full typeset reproduction therefore remains environment-dependent here, but the tracked repository surface is organized so the bounded claim surface, its imported reviewed downstream- boundary readout, and its support sources can be inspected without workspace- only links. Stripped comparison bundles are not the same thing as this tracked repository surface.

## 9 Discussion

The non-evaluative principle should enter this line as bounded engineering discipline rather than theorem inflation. In the current read, the practical issue is not simply whether unresolved content exists, but whether later interaction forces premature relation collapse while the interaction is still in a retained state. The `Guarantee` paper is the correct place to express that read as a bounded downstream claim surface. In that sense, the manuscript is adjacent to prior mixed-initiative, clarification, and grounding work, but its objective is different: it does not primarily propose a better question-asking heuristic or a general dialogue-success policy. Instead, it asks which bounded commit/defer claims remain supportable under explicit provider, family, horizon, and observability restrictions [8, 9, 11, 14, 15, 16, 17].

That downstream read should not be reduced to a simple rule of "never recommend one next move." A response may advance with a provisional recommendation and still preserve the user's live context if the governing constraint structure remains visible or explicitly marked as contingent. The harder failure is premature interpretation fixation: proceeding as if one local reading of the user's still-live objective, satisfaction condition, or constraint ordering had already been settled. Framed that way, unresolved-relation preservation is not mere hesitation. It is bounded context retention under a visible interaction state.

The current manuscript line also remains deliberately narrow. It is based on two tested provider slices, one supported early-control claim, one explicit downstream boundary, one side-topic supporting surface, and one later-horizon boundary read rather than on a broad cross-provider or long-horizon guarantee. Those limits are not defects to be hidden: they are part of the bounded-claim discipline that keeps the manuscript from overstating what the current evidence surface can support.

This staging matters for how the current line should be read. The manuscript is not claiming a monotone march from early control support to pair-wide downstream success. The truthful sequence is narrower: the early ambiguity-turn surface is clean on the tested provider pair, while the repaired downstream- acceptance boundary supports only bounded preacceptance-burden removal under matched `baseline vs Energy policy` conditions, while Gemini still shows `accepted_direction_followthrough_present` dropping from `13 / 24` to `10 / 24` and the auxiliary repaired `NRR vs Ablation` read remains mixed, with Anthropic still `inconclusive`, Gemini reaching `support`, and Anthropic `Baseline` remaining stronger than Anthropic `NRR` on `appropriate`. On that auxiliary repaired surface, the Anthropic gap is large rather than cosmetic: `Baseline` reaches `142 / 144` on `appropriate`, whereas `NRR` reaches `55 / 144`. That auxiliary asymmetry should remain visible because it marks a boundary on the repaired comparison surface, not because it upgrades the paper-level read. The paper-level downstream interpretation remains carried by the matched `baseline vs Energy policy` surface only: bounded support for preacceptance-burden removal is visible there, while pair-wide downstream- quality closure is not. Accordingly, the auxiliary repaired read is kept as boundary evidence of unresolved downstream fragility rather than as the authorizing basis for a paper-level early-control / downstream-quality trade-off claim.

The repaired prospective-killer pilot sharpens that boundedness rather than relaxing it. Its correct manuscript role is a secondary bridge note inside the evidence-import discussion: it shows that prompt-level `NRR-style` control can materially improve multi-turn behavior on the tested English bridge surface while still leaving later followthrough as the main residual family. But that bridge read should not be overread as a broader downstream-quality or mechanistic-closure result. The current evidence does not prove that `softmax` alone explains every remaining failure, the `Gemini-only logprobs` slice remains mechanistic sidecar evidence only, and the current package still falls short of establishing a non-softmax `NRR-native` architecture. Keeping those limitations adjacent to the reported pilot result

is part of the same bounded-claim discipline rather than an afterthought.

The present line should also be read as a reusable test surface rather than as the last word on guarantee behavior. What is fixed here is not an exhaustive operator set or a closed theorem, but a bounded way to keep later work honest: explicit supported claims, explicit supporting surfaces, explicit boundary surfaces, explicit comparison surfaces, provider-sliced reporting, and visible failure boundaries. That structure is meant to make follow-on work easier. Later readers should be able to rerun these guarantee questions under new providers, new downstream families, new operator variants, and longer horizons without having to guess which comparison contract or boundary rule the current manuscript relied on.

## 10 Limitations

The current manuscript is limited to a narrow tested-provider surface: two provider slices, one supported early-control claim, one explicit downstream boundary, one side-topic supporting surface, and one later-horizon boundary read rather than a broad cross-provider or long-horizon guarantee.

The downstream evidence also remains non-uniform. The repaired downstream- acceptance boundary supports bounded preacceptance-burden removal under matched `baseline vs Energy policy` conditions, but Gemini still shows reduced `accepted_direction_followthrough_present`, and the auxiliary repaired `NRR vs Ablation` surface remains mixed, with Anthropic still `inconclusive`. The prospective-killer pilot likewise remains secondary evidence only: it supports a bounded bridge read from prompt-level behavioral improvement toward future architectural motivation, but it does not establish pair-level continuation-quality superiority, provider-balanced mechanistic proof, or a non-softmax architecture claim. The current package should therefore be read as a reusable bounded test surface rather than as an exhaustive guarantee theorem or final cross-provider closure.

## 11 Conclusion

This manuscript establishes a bounded manuscript surface for `Guarantee`. It carries one supported claim unit for the early ambiguity-turn surface, carries the downstream-acceptance result forward as an explicit boundary surface with bounded support for `preacceptance_rework_burden_absent` on the mainline `baseline vs Energy policy` comparison while keeping the

repaired `NRR vs Ablation` read as an auxiliary comparator, carries the repaired first-step side-topic/quarantine result only as a bounded supporting surface under matched `baseline vs revised Energy policy` conditions, and keeps the later-horizon limit explicit as failure-boundary reporting rather than as a further supported claim. It also ties those supported and unsupported regions directly to the inherited policy vocabulary and dynamics motifs without promoting the line into provider-universal, family-universal, or theorem-level language. In that sense, the present `Guarantee` line is closer to a bounded verification/assurance surface over explicit observables than to a universal correctness claim [18, 19]. In practical terms, the current `Guarantee` line closes this pass around a staged bounded result: early control support does not automatically extend into pair-wide downstream acceptance, the repaired downstream boundary remains limited to burden removal rather than pair-wide downstream quality, and the Anthropic-side auxiliary comparison keeps `Baseline` stronger than `NRR` on `appropriate`. What remains outside the present claim surface is broader provider coverage, broader downstream family coverage, and later-horizon closure. The manuscript therefore aims to do one more thing than simply close the current bounded read: it fixes a reusable test surface that later readers can rerun, challenge, and extend under new conditions rather than treating the present line as exhaustive.

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| **Surface** | **Anthropic-side read** | **Gemini-side read** | **Boundary implication** |
|---|---|---|---|
| Early ambiguity-turn | Clean support on the tested early surface | Clean support on the tested early surface | Safe to state only a bounded early-control claim, not broader downstream closure |
| Downstream acceptance boundary | Mainline burden-removal support is clean; auxiliary repaired read stays `inconclusive` and keeps `Baseline` stronger than `NRR` on `appropriate` | Mainline burden-removal support is present, but `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`; auxiliary repaired read reaches `support` | Keep the claim at bounded burden-removal support under the mainline contract; use the repaired read only as boundary evidence, not as a replacement comparison rule |
| First-step side-topic / quarantine | Lift is visible, with strongest incremental gain on `C_incident_response` | Lift is also visible, again with the main gain concentrated on `C_incident_response` | Treat the result as template-bounded improvement, not uniform side-topic stability |
| Later-horizon boundary | Degradation remains visible at later checkpoints | Degradation remains visible at later checkpoints | Preserve later-horizon text as boundary reporting rather than a fourth supported guarantee |

Table 2: Provider-specific residual structure that keeps the current `Guarantee` claims bounded.